

LAWS OF MOTION

Dynamics of Particles, Force Fields, and Friction Modules

Subject: Theoretical Mechanics & Classical Dynamics

Level: Advanced Undergraduate / Competitive Physics Standard

Scope: Newton's Laws, Inertia, Linear Momentum, Linear Systems, Friction, and Circular Dynamics

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1. CLASSICAL DYNAMICS AND THE CONCEPT OF FORCE

While kinematics strictly isolates the geometric parameters of motion, **Dynamics** seeks to establish the qualitative and quantitative principles governing the physical causes of motion. The baseline entity translating these states is the intuitive and rigorous concept of Force.

1.1 Definition and Mathematical Character of Force

A force is a vector entity representing the mechanical interaction between two physical bodies or between a body and its environment. It acts to alter the state of absolute rest or uniform rectilinear translation of a target object. Being a vector quantity, a force possesses both an absolute scalar magnitude and an explicit spatial orientation:

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

1.2 Fundamental Forces of Nature

In classical and modern physics, all mechanical interactions are macroscopically and microscopically expressions of four fundamental force fields:

- **Gravitational Force:** An attractive force acting between particles due to their mass characteristics. It is long-range but inherently the weakest force field.
- **Electromagnetic Force:** Forces acting between electrically charged particles. It encompasses both static electrical forces and magnetic interactions, dominating macroscopic mechanical properties (such as tension, normal reactions, and friction).
- **Strong Nuclear Force:** Short-range, powerful force holding nucleons together within atomic nuclei.
- **Weak Nuclear Force:** Responsible for radioactive beta decay mechanisms and short-range nuclear transformations.

2. NEWTON'S THREE LAWS OF CLASSICAL MOTION

Sir Isaac Newton formulated three fundamental laws that serve as the axioms of classical mechanics, accurately detailing how force vectors manipulate the momentum configurations of mass matrices.

2.1 The First Law of Motion (Law of Inertia)

Newton's First Law states: *Every material body continues in its state of absolute rest or of uniform translation in a straight line unless it is compelled to change that state by external force vectors impressed upon it.*

This law fundamentally introduces the concept of **Inertia**, which is the inherent resistance of any physical mass to changes in its state of motion. Mass acts as the quantitative scalar measure of a system's inertia.

2.2 The Second Law of Motion (Fundamental Law of Dynamics)

Newton's Second Law provides the quantitative measurement framework for force: *The time rate of change of linear momentum of a body is directly proportional to the net applied external force vector and takes place in the exact spatial direction in which the force vector acts.*

2.2.1 Mathematical Formulation and Kinetic Derivation

Let linear momentum $\vec{p}$ be defined as the vector product of a particle's invariant mass m and its instantaneous velocity vector $\vec{v}$:

$$\vec{p} = m\vec{v}$$

According to the explicit axiom statement:

$$\vec{F} = d\vec{p} / dt \quad \vec{F} = k \cdot (d\vec{p} / dt)$$

In standard SI units, the constant of proportionality k is defined as 1. Expanding the differential derivative:

$$\vec{F} = d(m\vec{v}) / dt$$

By employing the standard calculus product rule:

$$\vec{F} = m(d\vec{v} / dt) + \vec{v}(dm / dt)$$

Within classical frameworks where mass is strictly invariant ($dm/dt = 0$), the expression simplifies to the iconic form:

$$\vec{F} = m\vec{a}$$

2.3 The Third Law of Motion (Action-Reaction Axiom)

Newton's Third Law states: *To every action, there is always an equal and opposite reaction; the mutual forces of two bodies upon each other are always equal and directed to contrary parts.*

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Crucially, action and reaction force vectors always develop simultaneously on two completely distinct bodies, meaning they can never cancel each other out within a multi-body equilibrium calculation.

[DIAGRAM 2.1: Interaction vector pairs between Contacting Block A and Block B, highlighting explicit Action Vector F_{AB} pushing Block B and the equivalent balancing Reaction Vector F_{BA} operating concurrently on Block A]

3. MECHANICS OF LINEAR IMPULSE AND MOMENTUM CONSERVATION

Analyzing short-duration, high-magnitude force transformations requires the introduction of integrated temporal variables.

3.1 Linear Impulse Vector Derivation

Impulse ($\vec{J}$) measures the net temporal integration of a force vector acting across a finite time window. From Newton's second law:

$$\vec{F} = d\vec{p} / dt \quad d\vec{p} = \vec{F} dt$$

Integrating across explicit initial and final momentum thresholds:

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt = \int_{p_1}^{p_2} d\vec{p} = \vec{p}_2 - \vec{p}_1 = \Delta\vec{p}$$

This derivation establishes the **Impulse-Momentum Theorem**, proving that the net vector impulse delivered equates identically to the linear momentum shift of the particle.

3.2 The Principle of Conservation of Linear Momentum

For an isolated multi-body system experiencing zero net external force vectors ($\vec{F}_{ext} = 0$):

$$d\vec{p}_{system} / dt = 0 \quad \vec{p}_{system} = constant$$

Thus, the total initial momentum configuration vector matches the final vector arrangement identically, serving as the base calculation rule for rocket propulsion mechanics and complex elastic collisions.

4. ANALYTICAL APPLICATION: FREE-BODY DIAGRAMS (FBD) AND CONSTRAINT EQUILIBRIUM

To solve macroscopically complex mechanics systems, one must computationally isolate target bodies via **Free-Body Diagrams**.

4.1 Protocol for Free-Body Diagram Disassembly

1. Isolate the specific body under mathematical review from its physical contacts and surrounds.
2. Represent the isolated body as a centralized point mass placed at its true center of gravity.
3. Map all active external vectors acting directly on the target body (Weight downwards, tension along strings, normal lines perpendicular to surfaces, friction opposing sliding trends).

4.2 Connected Mass Systems on Smooth Planes

Problem Profile: Two contact blocks (m_1 and m_2) propelled on a horizontal smooth track by force F

Since the blocks are structurally constrained to move in unison, they track with a singular common system acceleration a . Analyzing the consolidated system:

$$F = (m_1 + m_2)a \quad a = F / (m_1 + m_2)$$

To discover the internal normal contact force vector (N_c) shared between the blocks, we look closely at the independent FBD of mass m_2 :

$$N_c = m_2 a = m_2 F / (m_1 + m_2)$$

[DIAGRAM 4.1: Interconnected Mass Pulley System (Atwood Machine) displaying tension vector assignments (T) running along un-stretchable lines counterbalancing weight configurations ($m_1 \cdot g$) and ($m_2 \cdot g$)]

5. THE MECHANICS OF MACROSCOPIC FRICTION

Friction is a non-conservative contact force field that acts parallel to the interface surfaces, opposing any relative sliding motion between them. It originates from electromagnetic microscopic interlocking of surface irregularities.

5.1 Static Friction vs Kinetic Friction

- **Static Friction (f_s):** A self-adjusting balancing force that matches the magnitude of tangential pulling parameters until it reaches a maximum threshold known as limiting static friction:

$$f_{s,max} = \mu_s N$$

Where μ indicates the static coefficient of friction and N indicates the normal force reaction vector. s

- **Kinetic Friction (f_k):** The real sliding resistance force that develops once the static threshold is breached. It remains constant during uniform translation:

$$f_k = \mu_k N$$

Typically, $\mu_s > \mu_k$ due to fewer microscopic interlocking links being established during high-speed k sliding tracking.

5.2 Mechanics on an Inclined Plane Matrix

Consider a block of mass m sitting on a rough surface inclined at an angle θ relative to the ground. Resolving weight vectors into orthogonal axes yields components parallel ($mg \sin\theta$) and perpendicular ($mg \cos\theta$) to the incline surface.

$$N = mg \cos\theta$$
$$f_{s,max} = \mu_s mg \cos\theta$$

The system will begin to slide only when the downward parallel weight component surpasses the limiting friction threshold:

$$mg \sin\theta > \mu_s mg \cos\theta \quad \tan\theta > \mu_s$$

This specific boundary parameter defines the ****Angle of Repose**** ($\alpha = \tan^{-1} \mu_s$).

6. DYNAMICS OF UNIFORM AND NON-UNIFORM CIRCULAR MOTION

Applying Newton's second law to circular paths requires a complete transition into polar coordinate references, introducing centripetal force parameters.

6.1 Mathematical Definition of Centripetal Force

For a particle to follow a circular path of radius r , a net real force vector must act continuously inward toward the center of curvature to change its velocity direction. This real force requirement is defined as the Centripetal Force:

$$F_c = m \cdot a_c = mv^2 / r = m\omega^2 r$$

6.2 Banking of Curved Roadways

To reduce dependency on unstable tire friction along high-speed highway curves, engineers raise the outer edge of the roadway relative to the inner edge, creating an inclination angle θ known as ****Banking****.

[DIAGRAM 6.1: Cross-sectional FBD tracking of a vehicle traversing a banked roadway of angle theta. Normal force N is split into a vertical component ($N \cos \theta$) balancing weight and a horizontal component ($N \sin \theta$) providing safe centripetal force]

Resolving the normal reaction force N into vertical and horizontal components on a perfectly smooth banked road yields:

$$\begin{aligned} N \cos \theta &= mg \\ N \sin \theta &= mv^2 / r \end{aligned}$$

Dividing the equations provides the optimal design speed for a banked curve, independent of

friction forces:

$$\tan\theta = v^2 / (rg) \quad v_{\text{optimum}} = \sqrt{(rg \tan\theta)}$$